Quantitative Research Lab: 100 Strategies Across SPY, Sectors, Semiconductors and ETFs

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Strategies defined: 100. Strategies executed: 100. Instruments backtested: 32. Strategy x instrument combinations (full history): 2938. Execution: signals at the close, fills at the next open, 10 bps per side, lagged T-bill cash, long-only, no leverage.

Summary of findings

Strongest strategy families. Ranked by median out-of-sample Sharpe improvement over buy-and-hold: F09 Relative strength and sector rotation: median Sharpe-B&H; -0.00 full history, -0.01 walk-forward OOS (48% of assets positive); F07 Volatility and risk management: median Sharpe-B&H; -0.04 full history, -0.19 walk-forward OOS (8% of assets positive); F08 Regime detection: median Sharpe-B&H; -0.12 full history, -0.32 walk-forward OOS (2% of assets positive); F04 Mean reversion: median Sharpe-B&H; -0.27 full history, -0.36 walk-forward OOS (7% of assets positive). A family is only called consistent if most assets are positive out of sample; see Family_Summary for the full table.

Most reliable instruments and sectors. Out-of-sample Sharpe of 'pick the best backtest each year' minus buy-and-hold: best INTC (+0.17), KLAC (+0.04), NVDA (-0.00), AMAT (-0.03), QQQ (-0.04), XLK (-0.04); worst XLC (-0.40), DIA (-0.42), SOXX (-0.43), XLP (-0.46). Selection beat holding on 6% of 32 instruments.

Does SMA10/30 show a meaningful advantage?. SPY full history: SMA10/30 CAGR 6.1%, Sharpe 0.35 (90% bootstrap CI 0.08 to 0.63; DSR 0.30; FDR q 1.00), max drawdown -29.7%; SMA20/50 6.6% / 0.38 / -31.2%; buy-and-hold 10.9% / 0.51 / -55.2%. Cross-asset test (H2): Not resolved - Drawdown shallower for 10/30 on 53% of 32 assets (median +0.013, p=0.860); sideways-regime excess return lower for 10/30 on 47% (median +0.186%/yr, p=0.860); Sharpe higher for 10/30 on 44%

Do semiconductors behave differently from broad ETFs?. H1 (Not resolved): Fast-slow Sharpe: semis mean -0.132 (n=14, 7% positive); broad mean -0.072 (n=7); difference -0.060; p=0.172. H4 (Not resolved): Momentum Sharpe minus B&H;: semis -0.113 (n=14), broad -0.121 (n=7), diff +0.008, p=0.856. In up-trend regimes momentum excess vs B&H;: semis -6.11%/yr, broad -2.43%/yr (negative = momentum lags holding during expansions).

Do faster exits / slower re-entries help?. H3 (Not resolved): F02_02 vs F02_01: Sharpe better on 44% (median -0.019), drawdown better on 72%; F02_10 vs F01_02: Sharpe better on 31% (median -0.042), drawdown better on 81%; F02_06 vs F01_06: Sharpe better on 16% (median -0.066), drawdown better on 69%.

Which strategies were overfit?. Largest median drop from training-window to next-year Sharpe: F05_05 (+0.18), F05_02 (+0.11), F05_06 (+0.10), F02_04 (+0.09), F10_04 (+0.09), F10_02 (+0.08), F05_04 (+0.08), F03_07 (+0.08). Re-optimising parameters inside each training window lowered out-of-sample Sharpe versus the pre-registered default in 51% of strategy x asset cases. H8 (Supported): Mean Spearman +0.119 over 657 asset-folds; top-decile retention 12% vs chance 10% (t-test vs chance p=0.000); 58% of IS top-decile land in the OOS upper half; best-IS strategy's median OOS percentile 58%.

Which findings survive forward validation and multiple-testing control?. Return differential vs buy-and-hold: 0 of 2911 strategy x asset tests survive Benjamini-Hochberg FDR at 10% (none). Hansen SPA rejects 'no strategy beats holding' on 0 of 32 assets for raw returns and on 1 for volatility-matched returns.

Gap to the perfect long-only trader. SPY: daily oracle (costless decision, net of 10 bps) CAGR 155.6%; cost-aware unlimited 158.8%; with at most 10 entries 23.4%; buy-and-hold 10.9%. Even a handful of perfectly timed trades dwarfs every realistic rule, showing that the realised edge of timing rules is a tiny fraction of the theoretical opportunity. The oracle is impossible to implement.

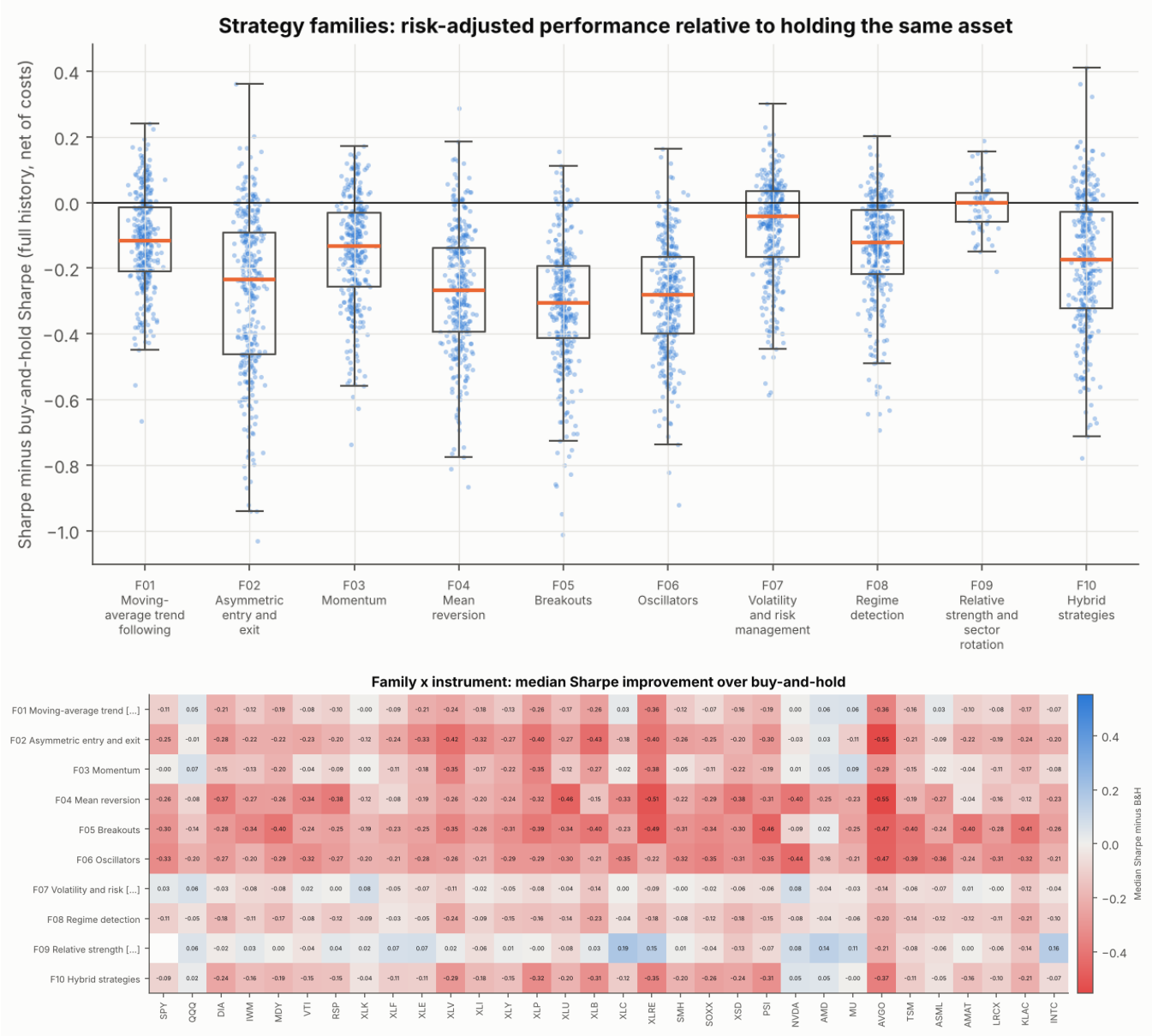
Remaining statistical uncertainty. Median width of 90% bootstrap Sharpe intervals: 0.60. Deflated Sharpe > 0.95 for 37 of 2911 strategy x asset cases (expected maximum Sharpe of ~45 effective trials: 0.40). Cross-asset tests treat correlated assets as independent and so overstate significance.

Research directions worth pursuing. Prospective paper-trading of the frozen registry from 2026-10-19 (the only truly untouched sample); point-in-time S&P; 500 constituents to remove survivorship bias; intraday execution data to calibrate slippage for volatile single stocks; ensemble construction across families with weights fixed ex ante; and regime models estimated strictly inside training windows.

Hypothesis tests (decision rules fixed before results; see Research_Methodology.md)

H	Claim	Verdict	Evidence
H1	Faster SMA systems work better than slower ones on volatile semiconductor assets	Not resolved	Fast-slow Sharpe: semis mean -0.132 (n=14, 7% positive); broad mean -0.072 (n=7); difference -0.060; p=0.172
H2	SMA10/30 offers better downside protection than SMA20/50 but lags in rapidly reversing (sideways) markets	Not resolved	Drawdown shallower for 10/30 on 53% of 32 assets (median +0.013, p=0.860); sideways-regime excess return lower for 10/30 on 47% (median +0.186%/yr, p=0.860); Sharpe higher for 10/30 on 44%
H3	Faster exits and slower re-entries improve risk-adjusted returns by limiting drawdowns	Not resolved	F02_02 vs F02_01: Sharpe better on 44% (median -0.019), drawdown better on 72%; F02_10 vs F01_02: Sharpe better on 31% (median -0.042), drawdown better on 81%; F02_06 vs F01_06: Sharpe better on 16% (median -0.066), drawdown better on 69%
H4	Semiconductor strategies benefit disproportionately from momentum during sustained industry expansion	Not resolved	Momentum Sharpe minus B&H;: semis -0.113 (n=14), broad -0.121 (n=7), diff +0.008, p=0.856. In up-trend regimes momentum excess vs B&H;: semis -6.11%/yr, broad -2.43%/yr (negative = momentum lags holding during expansions)
H5	Mean reversion performs better in sideways regimes than in strong directional trends	Contradicted	Sideways minus trend-regime Sharpe: median -0.164; positive for 25% of 32 assets (p=0.007); excess return vs B&H; sideways minus trend: median +2.61%/yr
H6	Sector-specific strategy selection outperforms one common signal for all sectors, after model-selection bias	Contradicted	OOS Sharpe specific minus common: mean -0.205 over 196 sector-folds (37% positive); fold-mean t-test p=0.045. Specific-selection IS->OOS Sharpe decay +0.66
H7	Performance deteriorates materially once transaction costs and slippage are included	Supported	0 bps: median Sharpe 0.37, 24% beat B&H; 5 bps: median Sharpe 0.33, 20% beat B&H; 10 bps: median Sharpe 0.30, 16% beat B&H; 20 bps: median Sharpe 0.24, 11% beat B&H; 50 bps: median Sharpe 0.09, 6% beat B&H.; Median Sharpe drop 0->10 bps 0.048; 90th pct 0.163
H8	Many top-ranked in-sample strategies fail to retain their ranking out of sample	Supported	Mean Spearman +0.119 over 657 asset-folds; top-decile retention 12% vs chance 10% (t-test vs chance p=0.000); 58% of IS top-decile land in the OOS upper half; best-IS strategy's median OOS percentile 58%
H9	Some apparent advantages disappear after matching invested exposure or volatility	Supported	2702 of 2911 combos have shallower drawdowns than B&H; vs the volatility-matched mix 15% have higher CAGR, 58% shallower drawdown, 14% both. Of 188 combos beating B&H; CAGR, 99% also beat the exposure-matched mix
H10	Combining complementary signals produces greater robustness than optimising a single indicator	Contradicted	share_pos_sharpe_diff: hybrids 0.209 vs median single-family 0.156 (better); disp_sharpe_diff: hybrids 0.203 vs median single-family 0.167 (worse); median_is_oos_decay: hybrids 0.026 vs median single-family -0.028 (worse); median_param_sharpe_std: hybrids 0.063 vs median single-family 0.059 (worse). Ensemble F10_07 beats B&H; Sharpe on 16% of assets

Strategy family comparison



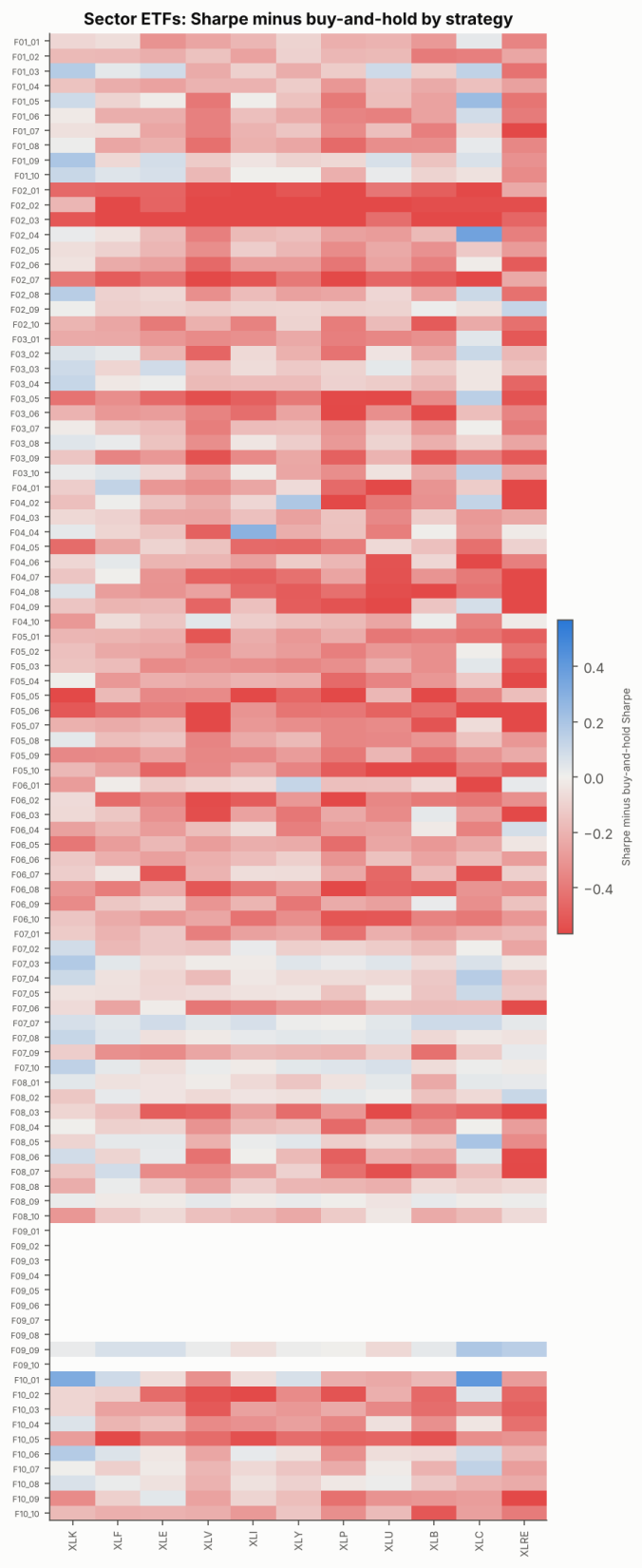
Leaderboard (top 15 by walk-forward out-of-sample Sharpe improvement, not by CAGR)

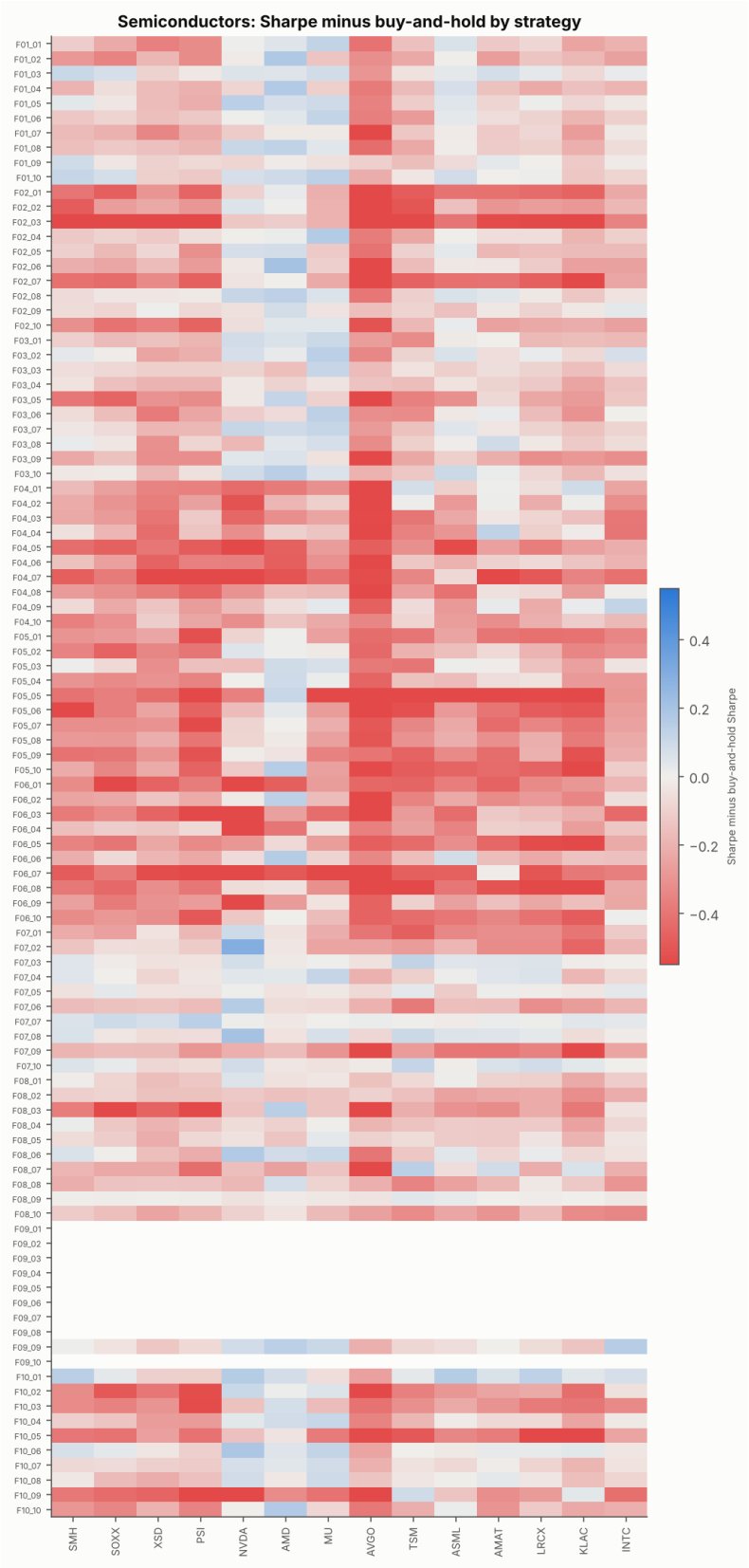
Asset	Strategy	Sharpe [90% CI]	B&H; Sharpe	CAGR	MaxDD	WF OOS S harpe-B&H;	DSR	FDR q
XLRE	F06_07 RSI bullish divergence proxy	0.21 [-0.10, 0.89]	0.32	4.2%	-31.6%	2.05	0.18	1.00
XLRE	F06_01 RSI(14) cross above 30, exit > 70	0.34 [-0.03, 0.81]	0.32	6.8%	-36.0%	1.19	0.30	1.00
XLC	F04_08 Three consecutive down closes	0.11 [-0.37, 0.62]	0.52	3.4%	-15.2%	1.06	0.12	1.00
XLC	F06_03 RSI(14) < 35 in uptrend (SMA200)	0.25 [-0.25, 1.12]	0.52	4.2%	-12.7%	0.88	0.22	1.00
XLC	F04_05 5% below SMA50 reversion	0.09 [-0.24, 0.53]	0.52	3.2%	-24.9%	0.74	0.12	1.00
XLRE	F04_05 5% below SMA50 reversion	0.22 [-0.12, 0.66]	0.32	4.7%	-31.6%	0.53	0.18	1.00
XLRE	F04_10 Buy 15% drawdown, sell near prior	0.31 [-0.07, 0.75]	0.32	6.3%	-27.4%	0.52	0.26	1.00
XLC	F09_09 Leadership switch: asset vs SPY (m	0.71 [0.22, 1.30]	0.52	16.5%	-30.2%	0.51	0.68	1.00
XLC	F06_07 RSI bullish divergence proxy	-0.00 [-0.26, 0.39]	0.52	2.4%	-18.4%	0.46	0.07	1.00
XLRE	F06_04 Stochastic %K/%D bullish cross bel	0.39 [-0.02, 1.04]	0.32	7.5%	-36.0%	0.31	0.36	1.00
XLI	F04_04 Bollinger reversion with SMA200 tr	0.70 [0.46, 0.92]	0.41	6.4%	-9.7%	0.26	0.92	1.00
XLRE	F09_09 Leadership switch: asset vs SPY (m	0.47 [0.05, 1.01]	0.32	9.9%	-39.5%	0.24	0.45	1.00
XLE	F09_09 Leadership switch: asset vs SPY (m	0.44 [0.16, 0.74]	0.37	10.1%	-54.8%	0.23	0.69	1.00
XLRE	F06_05 MACD line above signal	0.29 [-0.18, 0.72]	0.32	5.5%	-17.9%	0.23	0.23	1.00
MU	F06_04 Stochastic %K/%D bullish cross bel	0.46 [0.18, 0.75]	0.51	13.6%	-73.0%	0.22	0.68	1.00

Sharpe CIs: stationary bootstrap (mean block 20 days). DSR: deflated Sharpe ratio given the number of strategies tried on the asset. FDR q: Benjamini-Hochberg adjusted p-value for the return differential versus buy-and-hold across all strategy x asset tests.

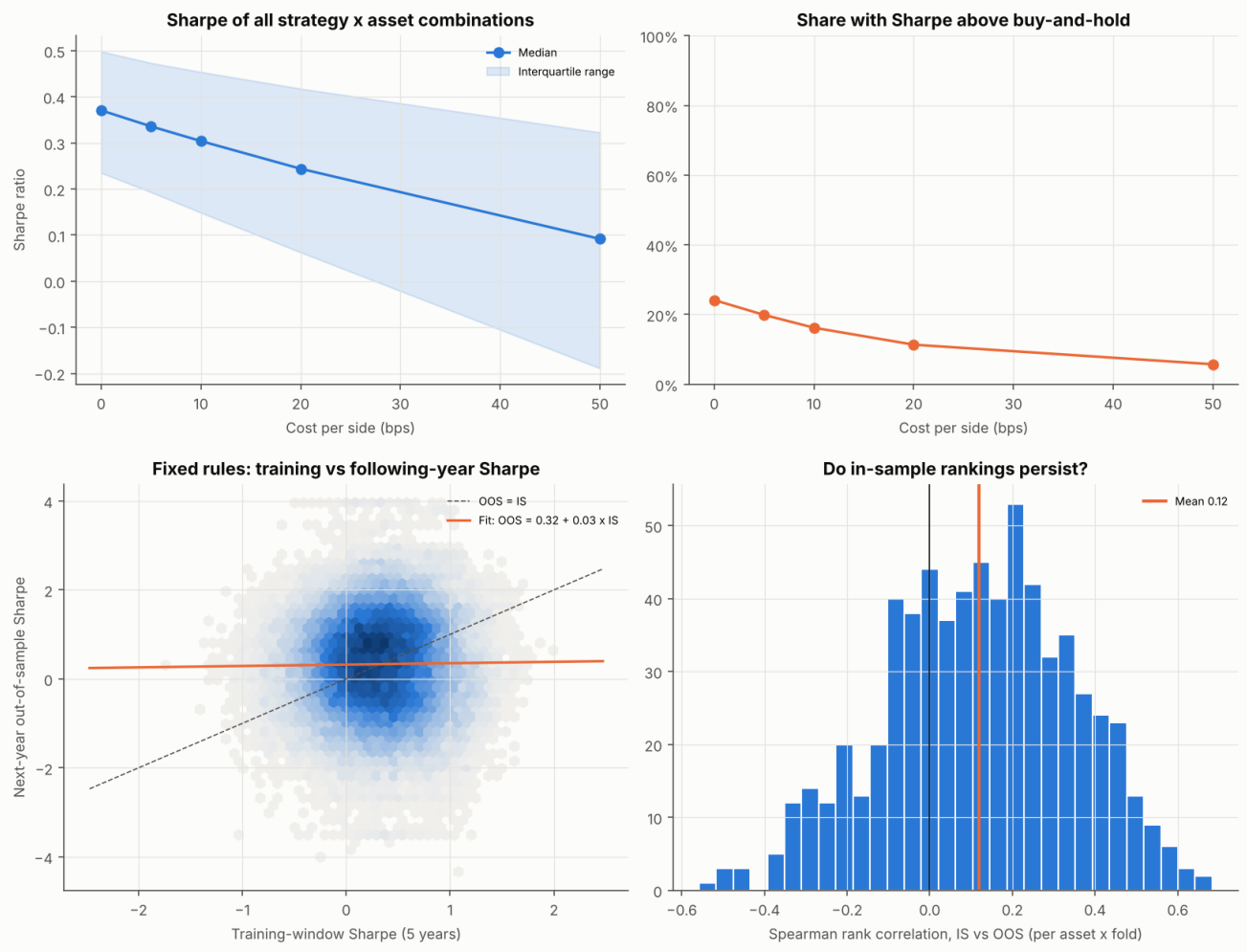

CAGR versus maximum drawdown (full available history, 10 bps per side)
Broad ETFs
Sector ETFs
Semi ETFs
Semi stocks
Strategy x asset
Buy-and-hold
CAGR (net)
Maximum drawdown
VTI
SPY
RSP
MDY
QQQ
IWM
DIA
XLC
XLK
XLE
XLY
XLI
XLB
XLU
XLV
XLP
XLF
XLRE
SMH
SOXX
XSD
PSI
AVGO
NVDA
ASML
TSM
KLAC
LRCX
AMAT
MU
AMD
INTC

Heatmaps

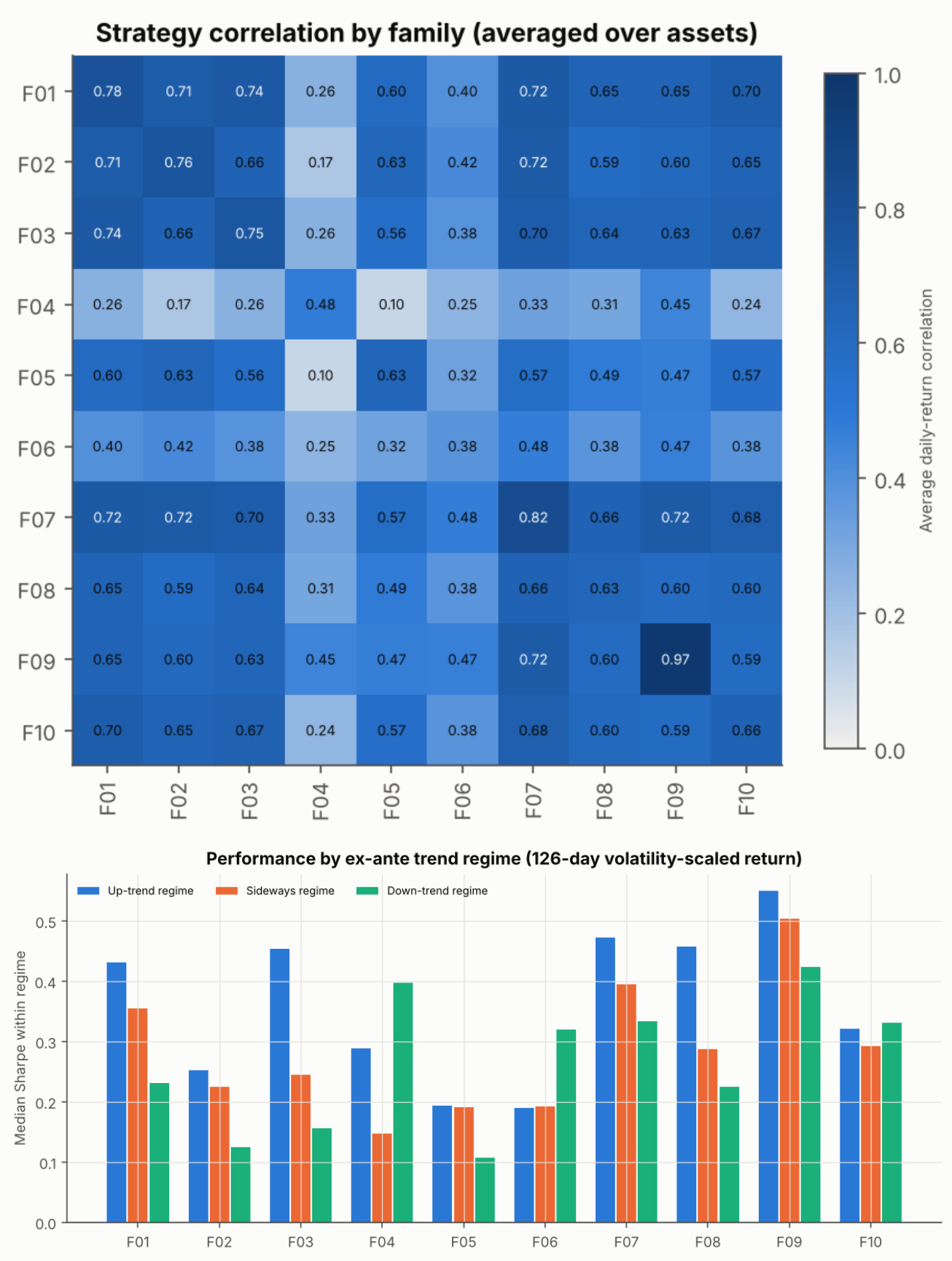

Sector ETFs: Sharpe minus buy-and-hold by strategy
XLK
XLF
XLE
XLV
XLI
XLY
XLP
XLU
XLB
XLC
XLRE
0.4
0.2
0.0
−0.2
−0.4
Sharpe minus buy-and-hold Sharpe


Semiconductors: Sharpe minus buy-and-hold by strategy
F01_01
F01_02
F01_03
F01_04
F01_05
F01_06
F01_07
F01_08
F01_09
F01_10
F02_01
F02_02
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F10_07
F10_08
F10_09
F10_10
SMH
SOXX
XSD
PSI
NVDA
AMD
MU
AVGO
TSM
ASML
AMAT
LRCX
KLAC
INTC
0.4
0.2
0.0
−0.2
−0.4
Sharpe minus buy-and-hold Sharpe

Robustness: costs, walk-forward, correlation and regimes




SPY: stitched out-of-sample record of picking the best backtest each year
Best-in-training strategy (walk-forward)
Buy-and-hold
Growth of $1 (log scale)
6 × 10^0
4 × 10^0
3 × 10^0
2 × 10^0
10^0
6 × 10^−1
Drawdown
0%
−10%
−20%
−30%
−40%
−50%
2000
2004
2008
2012
2016
2020
2024
2028


Strategy correlation by family (averaged over assets)
F01 0.78 0.71 0.74 0.26 0.60 0.40 0.72 0.65 0.65 0.70
F02 0.71 0.76 0.66 0.17 0.63 0.42 0.72 0.59 0.60 0.65
F03 0.74 0.66 0.75 0.26 0.56 0.38 0.70 0.64 0.63 0.67
F04 0.26 0.17 0.26 0.48 0.10 0.25 0.33 0.31 0.45 0.24
F05 0.60 0.63 0.56 0.10 0.63 0.32 0.57 0.49 0.47 0.57
F06 0.40 0.42 0.38 0.25 0.32 0.38 0.48 0.38 0.47 0.38
F07 0.72 0.72 0.70 0.33 0.57 0.48 0.82 0.66 0.72 0.68
F08 0.65 0.59 0.64 0.31 0.49 0.38 0.66 0.63 0.60 0.60
F09 0.65 0.60 0.63 0.45 0.47 0.47 0.72 0.60 0.97 0.59
F10 0.70 0.65 0.67 0.24 0.57 0.38 0.68 0.60 0.59 0.66
F01 F02 F03 F04 F05 F06 F07 F08 F09 F10
Average daily-return correlation
1.0 0.8 0.6 0.4 0.2 0.0
Performance by ex-ante trend regime (126-day volatility-scaled return)
Up-trend regime
Sideways regime
Down-trend regime
Median Sharpe within regime
0.5 0.4 0.3 0.2 0.1 0.0
F01 F02 F03 F04 F05 F06 F07 F08 F09 F10

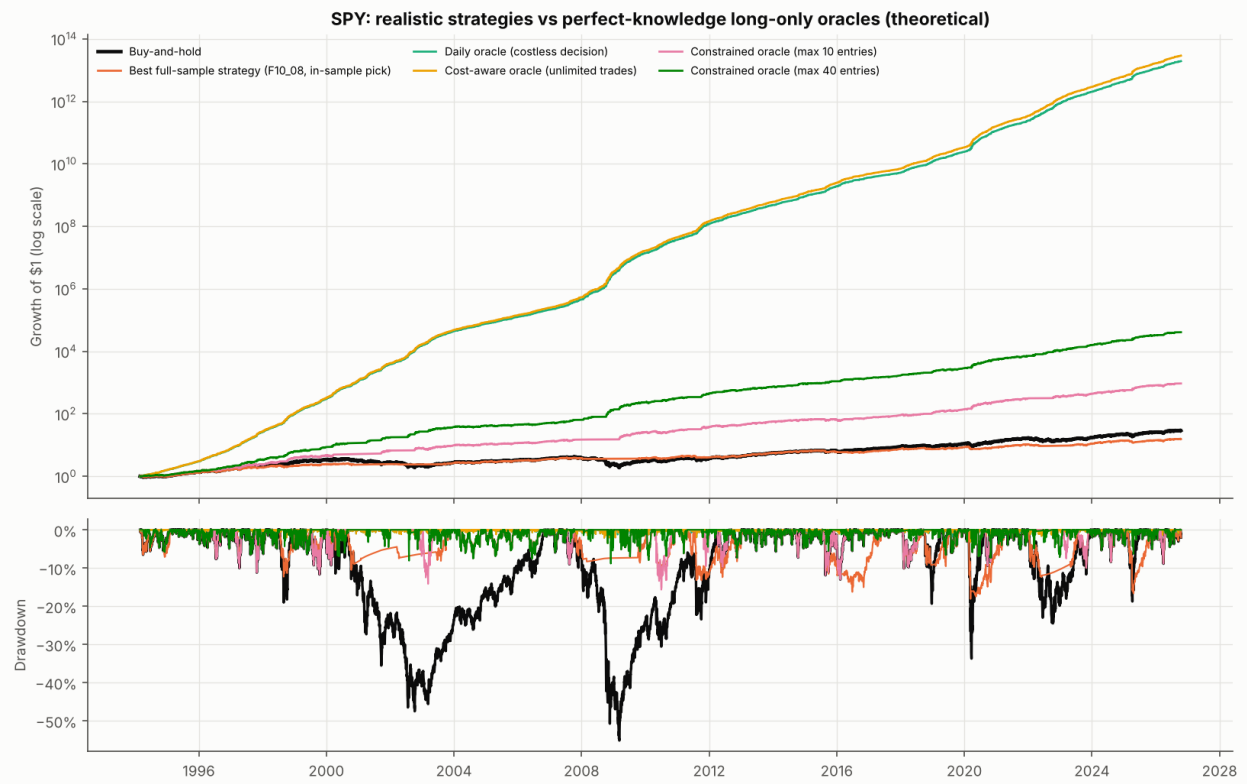
Multiple testing: Reality Check / SPA per instrument

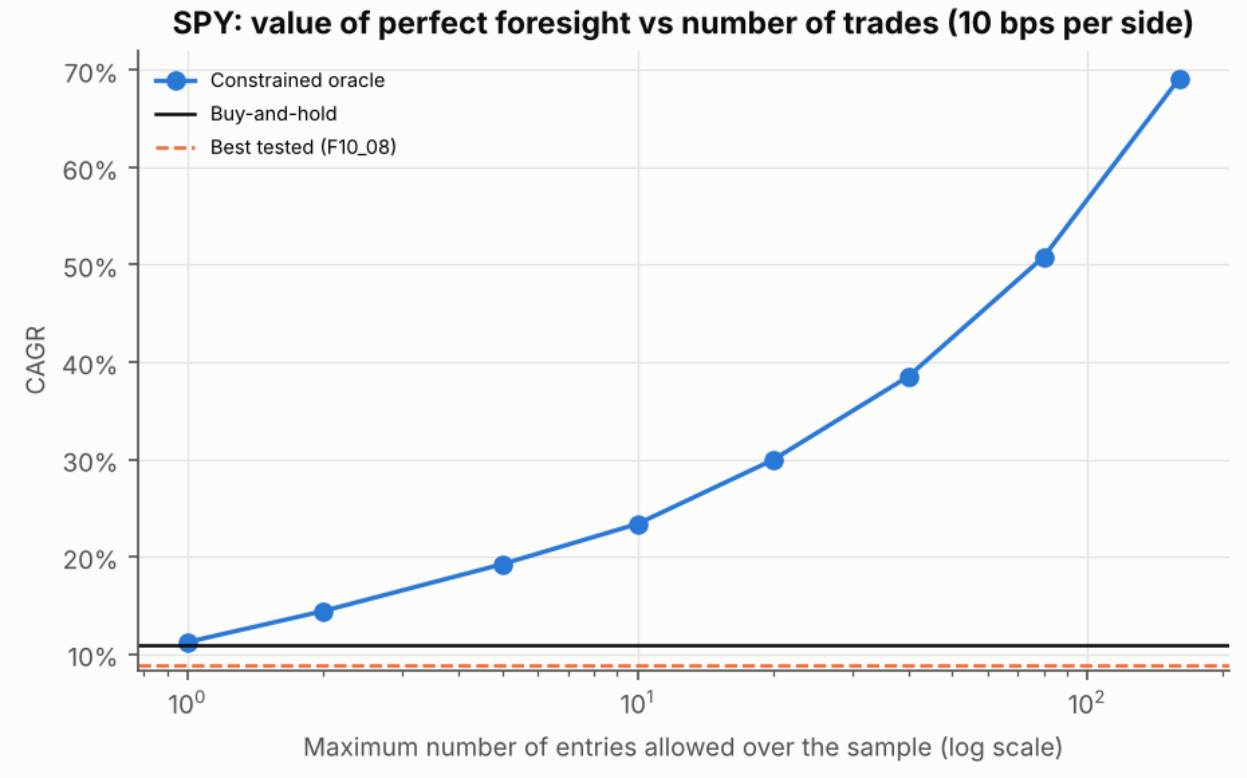
ticker	n_strategies	best_strategy	best_mean_diff_ann	rc_pvalue	spa_pvalue	best_strategy_vol_scaled	spa_pvalue_vol_scaled
IWM	91	F07_07	3.1%	0.653	0.487	F07_07	0.648
DIA	91	F07_07	0.0%	0.990	0.976	F07_07	0.960
QQQ	91	F10_01	4.2%	0.553	0.626	F10_01	0.070
SPY	90	F07_07	-0.2%	0.981	1.000	F10_08	0.913
RSP	91	F07_07	1.4%	0.838	0.650	F07_07	0.539
VTI	91	F07_07	0.5%	0.932	0.802	F07_03	0.589
MDY	91	F07_07	2.1%	0.739	0.460	F07_07	0.528
XLK	91	F10_01	3.7%	0.556	0.633	F10_01	0.170
XLF	91	F07_07	0.6%	0.952	0.908	F04_01	0.957
XLE	91	F07_07	1.6%	0.874	0.764	F07_07	0.867
XLV	91	F09_09	0.3%	0.967	0.935	F07_07	0.962
XLI	91	F07_07	0.7%	0.920	0.857	F04_04	0.773
XLY	91	F08_09	-0.3%	0.987	1.000	F04_02	0.946
XLP	91	F09_09	0.3%	0.977	0.956	F07_10	0.931
XLU	91	F07_07	0.1%	0.985	0.969	F01_03	0.926
XLB	91	F07_07	1.7%	0.813	0.580	F07_07	0.649
XLC	91	F10_01	3.3%	0.811	0.771	F10_01	0.509
XLRE	91	F09_09	2.1%	0.880	0.845	F09_09	0.963
SMH	91	F07_07	1.2%	0.930	0.802	F10_01	0.915
SOXX	91	F07_07	1.8%	0.893	0.664	F07_07	0.770
XSD	91	F07_07	1.0%	0.949	0.846	F07_07	0.807
PSI	91	F07_07	3.2%	0.828	0.474	F07_07	0.453
NVDA	91	F10_01	-0.4%	0.990	1.000	F07_02	0.310
AMD	91	F02_09	1.4%	0.949	0.935	F02_06	0.741
AVGO	91	F07_07	-2.1%	1.000	1.000	F07_07	0.998
MU	91	F07_07	-0.6%	0.985	1.000	F02_04	0.845
TSM	91	F08_09	-0.6%	0.994	1.000	F08_07	0.630
ASML	91	F08_09	-0.6%	0.993	1.000	F10_01	0.736
AMAT	91	F07_07	0.0%	0.987	0.963	F04_04	0.944
LRCX	91	F07_07	-0.3%	0.989	1.000	F10_01	0.869
KLAC	91	F07_07	0.9%	0.974	0.799	F04_01	0.900
INTC	91	F09_09	1.4%	0.923	0.880	F09_09	0.650

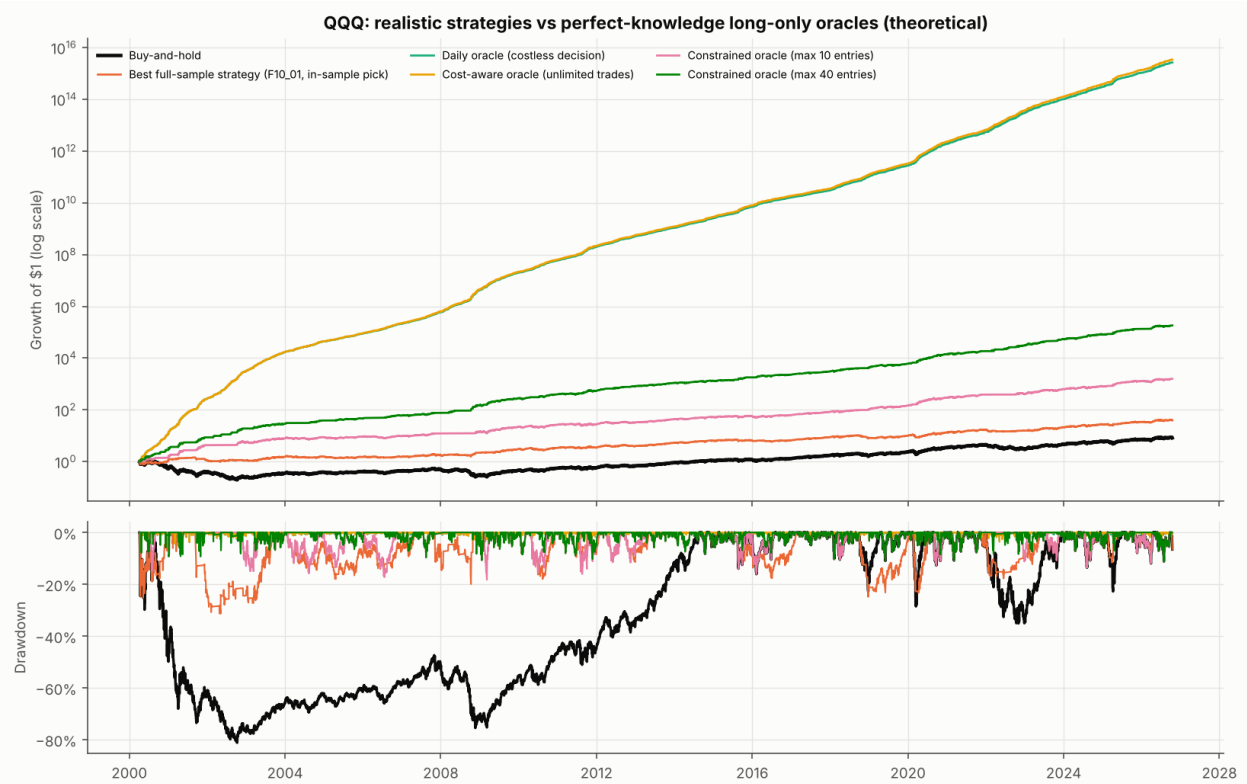
H0: no strategy has a positive expected daily return differential versus buy-and-hold of the same asset. The volatility-scaled version first scales each strategy's excess return to the asset's volatility, testing risk-adjusted rather than raw outperformance.

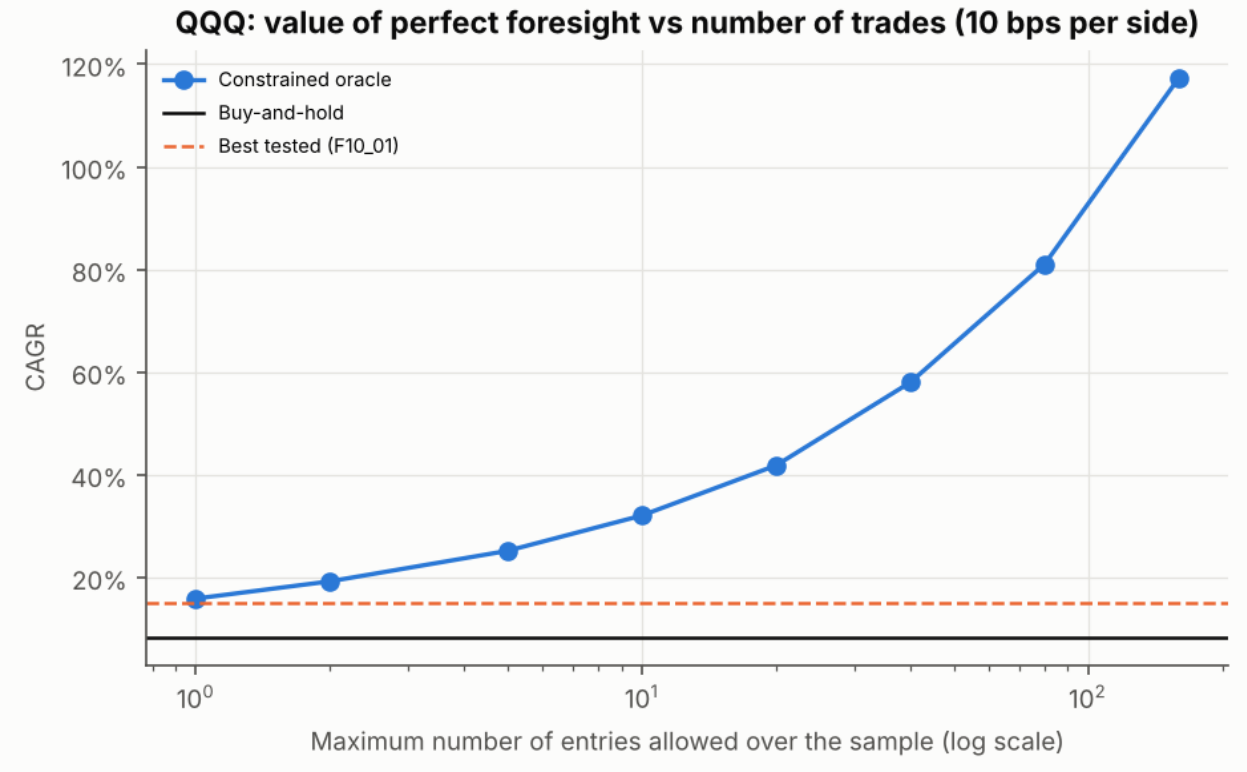
Perfect-knowledge long-only oracle (theoretical benchmark)

The oracle trades at the same opens as the strategies but knows the next open-to-open return. It is mathematically informative and impossible to implement.









SPY

benchmark	cagr	sharpe	max_dd	time_invested	entries	turnover_ann
Daily oracle (costless decision)	155.6%	7.42	-3.6%	54.2%	2095	128.5
Daily oracle (costless decision) [0 bps]	190.7%	8.47	-3.6%	54.2%	2095	128.5

benchmark	cagr	sharpe	max_dd	time_invested	entries	turnover_ann
Cost-aware oracle (unlimited trades)	158.8%	7.53	-3.6%	55.8%	1705	104.6
Constrained oracle (max 1 entries)	11.2%	0.53	-55.2%	99.2%	1	0.1
Constrained oracle (max 2 entries)	14.4%	0.82	-33.7%	71.8%	2	0.1
Constrained oracle (max 5 entries)	19.2%	1.11	-19.3%	84.5%	5	0.3
Constrained oracle (max 10 entries)	23.4%	1.41	-15.7%	81.9%	10	0.6
Constrained oracle (max 20 entries)	29.9%	1.72	-11.7%	80.3%	20	1.2
Constrained oracle (max 40 entries)	38.5%	2.21	-8.9%	77.2%	40	2.5
Constrained oracle (max 80 entries)	50.8%	2.83	-6.3%	73.6%	80	4.9
Constrained oracle (max 160 entries)	69.1%	3.82	-4.7%	68.3%	160	9.8

QQQ

benchmark	cagr	sharpe	max_dd	time_invested	entries	turnover_ann
Daily oracle (costless decision)	282.4%	7.55	-5.1%	54.3%	1697	128.1
Daily oracle (costless decision) [0 bps]	334.7%	8.28	-4.9%	54.3%	1697	128.2
Cost-aware oracle (unlimited trades)	286.1%	7.59	-5.1%	55.2%	1460	110.2
Constrained oracle (max 1 entries)	15.9%	0.72	-53.4%	90.5%	1	0.1
Constrained oracle (max 2 entries)	19.2%	0.91	-35.1%	86.5%	2	0.2
Constrained oracle (max 5 entries)	25.2%	1.15	-28.6%	84.4%	5	0.4
Constrained oracle (max 10 entries)	32.1%	1.42	-18.3%	83.9%	10	0.8
Constrained oracle (max 20 entries)	41.8%	1.81	-16.1%	78.3%	20	1.5
Constrained oracle (max 40 entries)	58.0%	2.35	-11.2%	72.4%	40	3.0
Constrained oracle (max 80 entries)	81.1%	3.12	-8.3%	69.3%	80	6.0
Constrained oracle (max 160 entries)	117.3%	4.12	-6.1%	65.3%	160	12.1

SMH

benchmark	cagr	sharpe	max_dd	time_invested	entries	turnover_ann
Daily oracle (costless decision)	505.6%	8.20	-5.9%	52.6%	1629	129.0
Daily oracle (costless decision) [0 bps]	589.1%	8.79	-5.7%	52.6%	1629	129.1
Cost-aware oracle (unlimited trades)	510.2%	8.23	-5.9%	53.6%	1446	114.5
Constrained oracle (max 1 entries)	21.2%	0.83	-45.3%	69.5%	1	0.1
Constrained oracle (max 2 entries)	25.9%	0.92	-45.3%	74.5%	2	0.2
Constrained oracle (max 5 entries)	34.8%	1.21	-33.6%	70.1%	5	0.4
Constrained oracle (max 10 entries)	44.8%	1.54	-23.7%	69.6%	10	0.8
Constrained oracle (max 20 entries)	58.9%	1.86	-18.3%	74.0%	20	1.6
Constrained oracle (max 40 entries)	80.4%	2.38	-15.6%	68.7%	40	3.2
Constrained oracle (max 80 entries)	115.9%	3.22	-11.7%	65.2%	80	6.3
Constrained oracle (max 160 entries)	173.3%	4.26	-9.6%	62.1%	160	12.7

NVDA

benchmark	cagr	sharpe	max_dd	time_invested	entries	turnover_ann
Daily oracle (costless decision)	2422.1%	8.06	-7.3%	52.0%	1685	126.6
Daily oracle (costless decision) [0 bps]	2762.7%	8.37	-7.2%	52.0%	1685	126.7

benchmark	cagr	sharpe	max_dd	time_invested	entries	turnover_ann
Cost-aware oracle (unlimited trades)	2433.8%	8.07	-7.3%	52.3%	1567	117.7
Constrained oracle (max 1 entries)	37.5%	0.86	-85.1%	90.0%	1	0.1
Constrained oracle (max 2 entries)	47.7%	1.04	-66.3%	86.2%	2	0.2
Constrained oracle (max 5 entries)	70.2%	1.26	-65.4%	87.7%	5	0.4
Constrained oracle (max 10 entries)	100.2%	1.68	-51.9%	77.4%	10	0.8
Constrained oracle (max 20 entries)	142.7%	2.16	-31.7%	76.1%	20	1.5
Constrained oracle (max 40 entries)	208.5%	2.68	-24.4%	72.2%	40	3.0
Constrained oracle (max 80 entries)	321.0%	3.40	-18.4%	67.4%	80	6.0
Constrained oracle (max 160 entries)	518.2%	4.33	-11.2%	62.7%	160	12.0

XLU

benchmark	cagr	sharpe	max_dd	time_invested	entries	turnover_ann
Daily oracle (costless decision)	167.3%	7.64	-5.4%	52.6%	1713	128.3
Daily oracle (costless decision) [0 bps]	204.0%	8.65	-5.2%	52.6%	1713	128.4
Cost-aware oracle (unlimited trades)	170.4%	7.75	-5.4%	53.6%	1404	105.2
Constrained oracle (max 1 entries)	10.9%	0.57	-46.5%	87.7%	1	0.1
Constrained oracle (max 2 entries)	13.6%	0.77	-36.1%	83.0%	2	0.1
Constrained oracle (max 5 entries)	18.4%	1.11	-17.6%	81.0%	5	0.4
Constrained oracle (max 10 entries)	23.0%	1.35	-15.7%	80.1%	10	0.7
Constrained oracle (max 20 entries)	29.7%	1.70	-12.0%	75.9%	20	1.5
Constrained oracle (max 40 entries)	40.2%	2.37	-8.6%	67.4%	40	3.0
Constrained oracle (max 80 entries)	54.9%	3.10	-5.6%	65.2%	80	6.0
Constrained oracle (max 160 entries)	76.6%	4.04	-5.3%	62.3%	160	12.0

Limitations

- Prices are Yahoo Finance daily bars adjusted locally for splits and distributions; they are research-grade convenience data, not exchange-quality.

- Execution assumes fills at the official open plus a fixed bps cost; opening-auction slippage and market impact for large orders are not modelled.

- Fractional shares and costless cash sweeps are assumed; taxes are ignored.

- Universe E (historical S&P; 500 constituents) was not run: no point-in-time membership and delisting data were available. No current-constituent study is presented as unbiased.

- The SPY sample 2010-2026 was inspected in the prior notebook and is exploratory. All historical data predate the registry freeze only in the sense that strategy ideas were drawn from public literature; the only genuinely untouched sample is prospective (after 2026-10-18).

- Cross-asset statistics treat correlated instruments as independent and therefore overstate significance.

- Semiconductor ETF histories differ in construction over time (e.g. SMH's pre-2011 HOLDRS history).